检测报告

编号 : NBIN2101000066PS_CN

日期 : 2021-01-20

扫码查看在线报告

NBIN2101000066PS

客户名称： 宁波哲能精密塑料有限公司

客户地址： 宁波市江北区姜湖路 406 号 5 幢 6 幢

样品名称 : PPS 板棒

以上信息及样品由客户提供及确认，SGS 不承担证实客户提供信息的准确性、适当性和(或)完整性的责任。

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收样日期 : 2021-01-04

测试开始时间 : 2021-01-04

测试结束时间 : 2021-01-20

测试结果 : 请见下页（除另有特别说明外，此报告结果仅对测试样品负责）

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严波 授权签字人

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结果总结：

序号	测试项目	测试方法	结果	结论
1	拉伸测试	ISO 527-1:2019 & ISO 527-2:2012	见测试结果	/
2	弯曲测试	ISO 178:2019 方法 A	见测试结果	/
3	简支梁缺口冲击强度	ISO 179-1:2010	2.7 kJ/m^2 （C-完全破坏）	/
4	洛氏硬度	ISO 2039-2:1987	101	/
5	比重	ISO 1183-1:2019 方法 A	1.366	/
6	吸水率	ISO 62:2008 方法 1	0.049%	/
7	线性热膨胀系数	ISO 11359-1:2014 & ISO 11359-2:1999 方法 A	114×10^{-6} K^{-1}	/
8	负荷变形温度	ISO 75-1:2020 & ISO 75-2:2013 方法 A	201℃	/
9	熔融温度	ISO 11357-3:2018	见测试结果	/

备注： 合格：达到要求

不合格：未达到要求

/：不下判定

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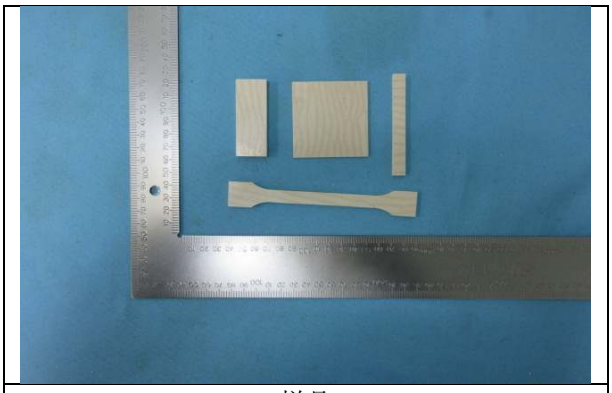
原样照片：



样品

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1. 测试项目：拉伸测试

样品描述：塑料样条

测试方法： ISO 527-1:2019 & ISO 527-2:2012

测试条件：

类型：1A

测试速度：10mm/min

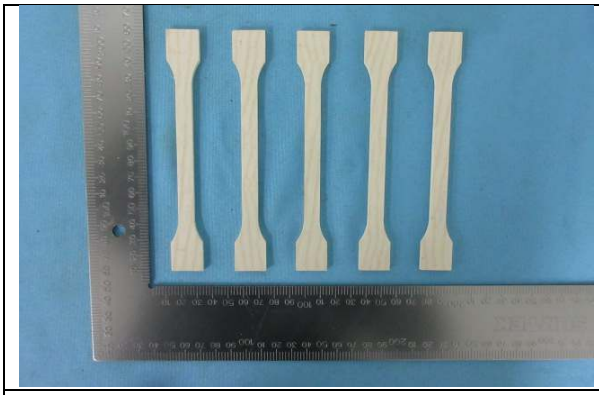
标距：75 mm

实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果	
拉伸测试	拉伸强度	65.2MPa
	断裂伸长率	5.2%

测试照片：



测试样

测试中

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
Zwick 电子万能材料试验机	Z010 TH Proline	NBPL-A025	2020-06-19	2021-06-18

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2. 测试项目：弯曲测试

样品描述：塑料样条

测试方法：ISO 178:2019 方法 A

测试条件：

测试速度：2mm/min

跨距：58mm

压头直径：10mm

支座直径：10mm

实验室环境条件：(23±2)℃，(50±5)%RH

测试结果：

测试项目	测试结果	
弯曲测试	在规定挠度时弯曲应力	102 MPa 注1
	最大弯曲应力	123 MPa
	弯曲模量	2880 MPa

注：该样品在规定挠度前即不屈服也不断裂，为标准 ISO 178:2019 中曲线 C，规定挠度即弯曲应变为 3.5%。

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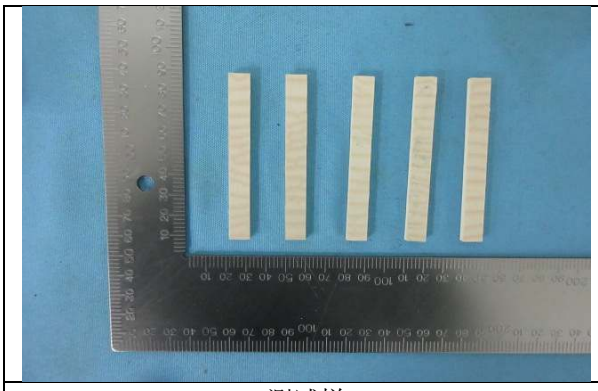	
测试样	测试中

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
Zwick 电子万能材料试验机	Z010 TH Proline	NBPL-A025	2020-06-19	2021-06-18

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3. 测试项目：简支梁缺口冲击强度

样品描述：塑料样条

测试方法：ISO 179-1:2010

测试条件：

试样：ISO 179-1/1eA (缺口制备：机械加工)

摆锤能量：1J

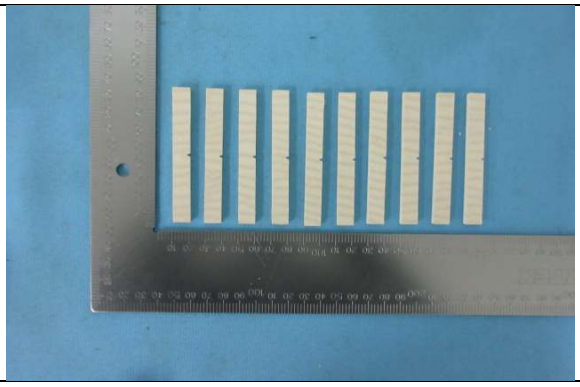
跨距：62mm

实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果
简支梁缺口冲击强度	2.7 kJ/m^2 (C-完全破坏)

测试照片：



测试样

测试中

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
摆锤式冲击仪	CEAST 9050	NBPL-A026	2020-02-24	2021-02-23

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4. 测试项目：洛氏硬度

样品描述：塑料样条

测试方法：ISO 2039-2:1987

测试条件：

试样厚度：6.79mm

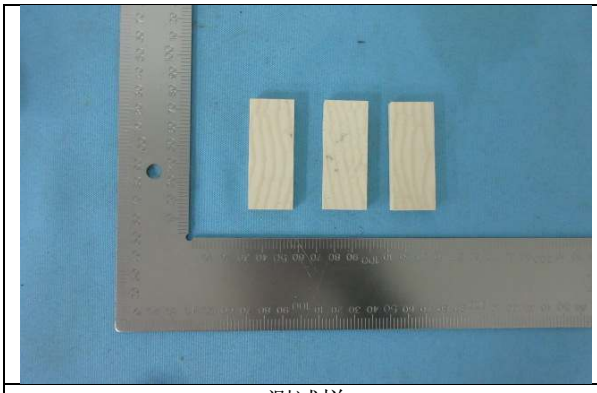
标尺类型：HRL

实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果
洛氏硬度	101

测试照片：



测试样

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
洛氏硬度计	660RLD-T	NBPL-A029	2020-08-03	2021-08-02

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5. 测试项目：比重

样品描述：塑料样条

测试方法：ISO 1183-1:2019 方法 A

测试条件：无水乙醇，23±2 ℃

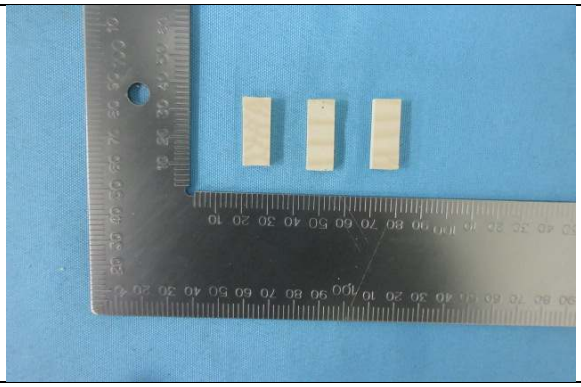
实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果
比重	1.369

注：测试样从样品中裁取

测试照片：



测试样

设备信息：

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密度天平	BS224S	NBPL-A030	2020-08-03	2021-08-02

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6. 测试项目：吸水率

样品描述：塑料样条

测试方法：ISO 62:2008 方法 1

测试条件：

浸泡条件：(23±2)℃，24h

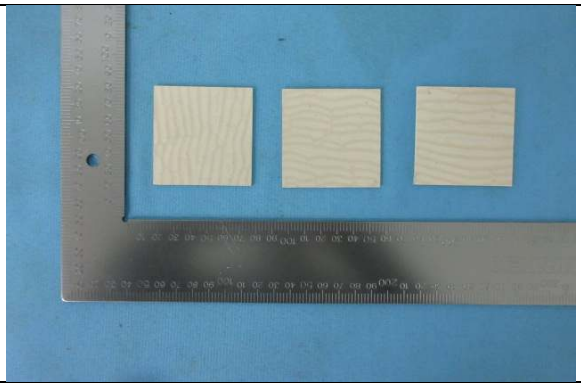
实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果
吸水率	0.049%[注]

注：吸水率，%=(浸泡后的质量-浸泡前干燥后的质量)/浸泡前干燥后的质量×100

测试照片：



测试样

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设备	型号	设备编号	校准日期	下次校准日期
密度天平	BS224S	NBPL-A030	2020-08-03	2021-08-02

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检测报告

7. 测试项目：线性热膨胀系数

样品描述：塑料样条

测试方法：ISO 11359-1:2014 & ISO 11359-2:1999 方法 A

测试条件：

试样：10.02mm×10.03mm×6.87mm

升温速率：5℃/min

负载：4kPa

气体流动速率(N_2)：50ml/min

测试温度：100℃~120℃

测量模式：压缩

测试方向：箭头

试样数：3

实验室环境条件：(23±2)℃, (50±5)%RH

测试结果：

测试项目	测试结果
线性热膨胀系数	114×10^{-6} K^{-1}

原始数据：

试样	#1	#2	#3
线性热膨胀系数，$\times10^{-6}$ K^{-1}	119.0	114.3	109.1

注： 测试样从样品中裁取。

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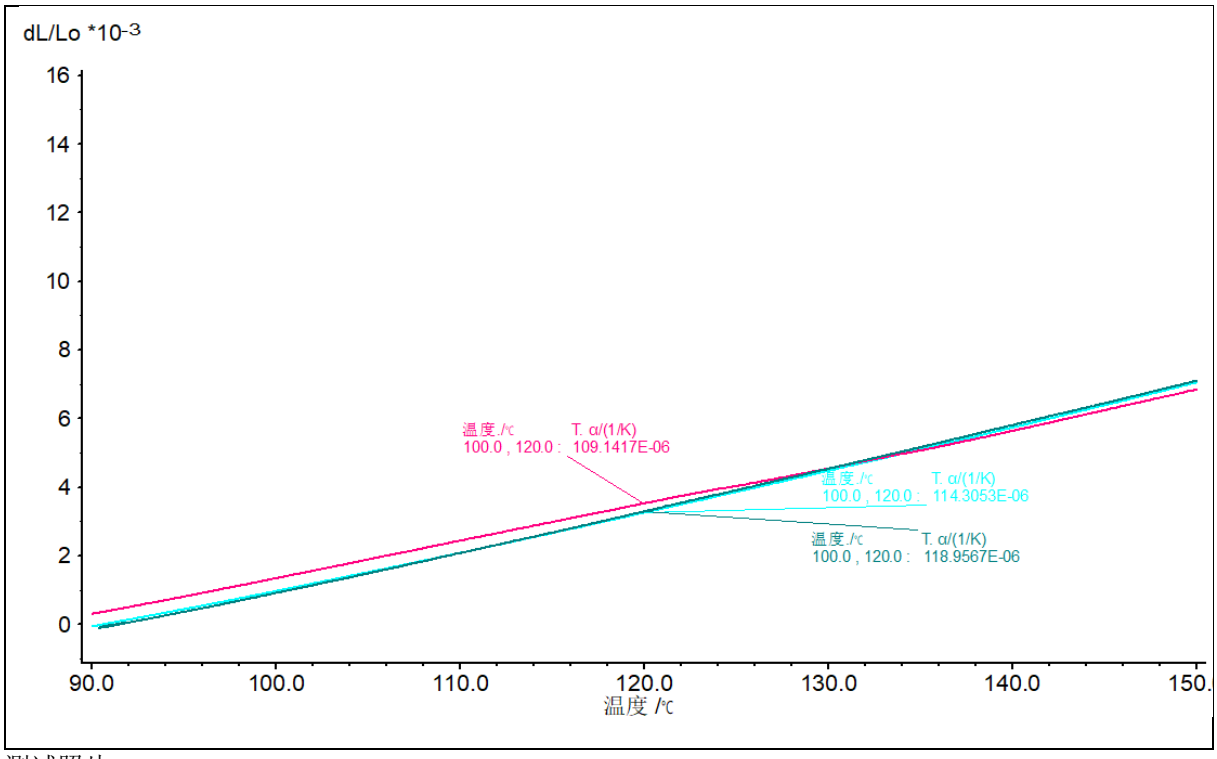
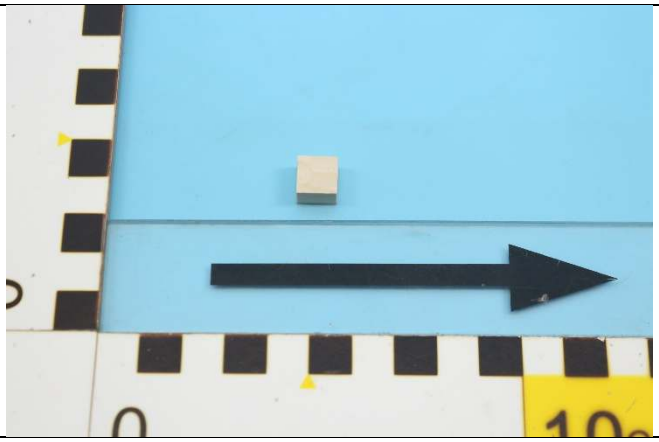
曲线图：



测试照片：



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	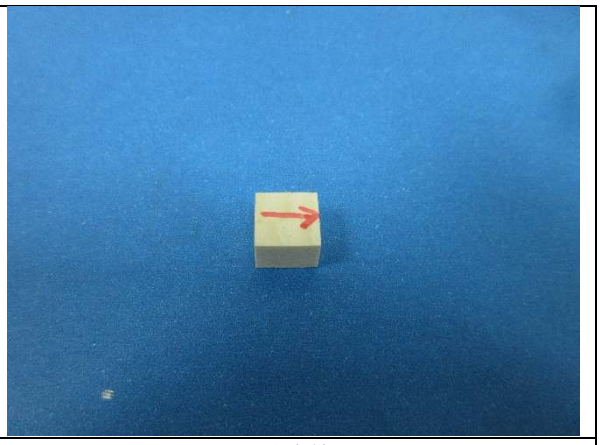
测试中	测试前

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
热机械分析仪	TMA402F3	GZMR-PL-E158	2020-07-15	2021-07-14

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8. 测试项目：负荷变形温度

样品描述：塑料样条

测试方法：ISO 75-1:2020 & ISO 75-2:2013 方法 A

测试条件：

试样：80mm×10.02mm×3.99mm

传热介质：硅油

T_f1.8

升温速率：120℃/h

跨距：64mm

实验室环境条件：23 ± 2 ℃，50 ± 5 % RH

测试结果：

测试项目	测试结果
负荷变形温度	201℃

原始数据：

试样	#1	#2	#3
负荷变形温度，℃	201.5	201.8	200.1

注：纹路面朝向负载。

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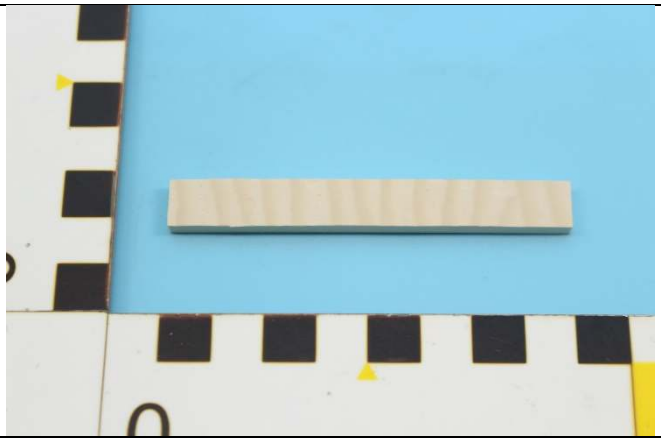

原样

测试中

测试后

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
热变形维卡测定仪	6970.000	GZMR-PL-E091	2020-02-26	2021-02-25

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9. 测试项目：熔融温度

样品描述：塑料样条

测试方法： ISO 11357-3:2018，分析仪器为 DSC。

测试条件：

样品容器类型：Al；

吹扫气体流速(氮气：99.999%)：50ml/min；

升温范围： 40~320℃；

升温速率： 10℃/min。

测试结果：

测试项目	第一次升温扫描 (DSC 曲线见图 1)	第二次升温扫描 (DSC 曲线见图 2)
外推熔融起始温度 T_{eim}	262.0℃	269.2℃
熔融峰温 T_{pm}	282.3℃	280.4℃
外推熔融终止温度 T_{efm}	287.3℃	285.0℃

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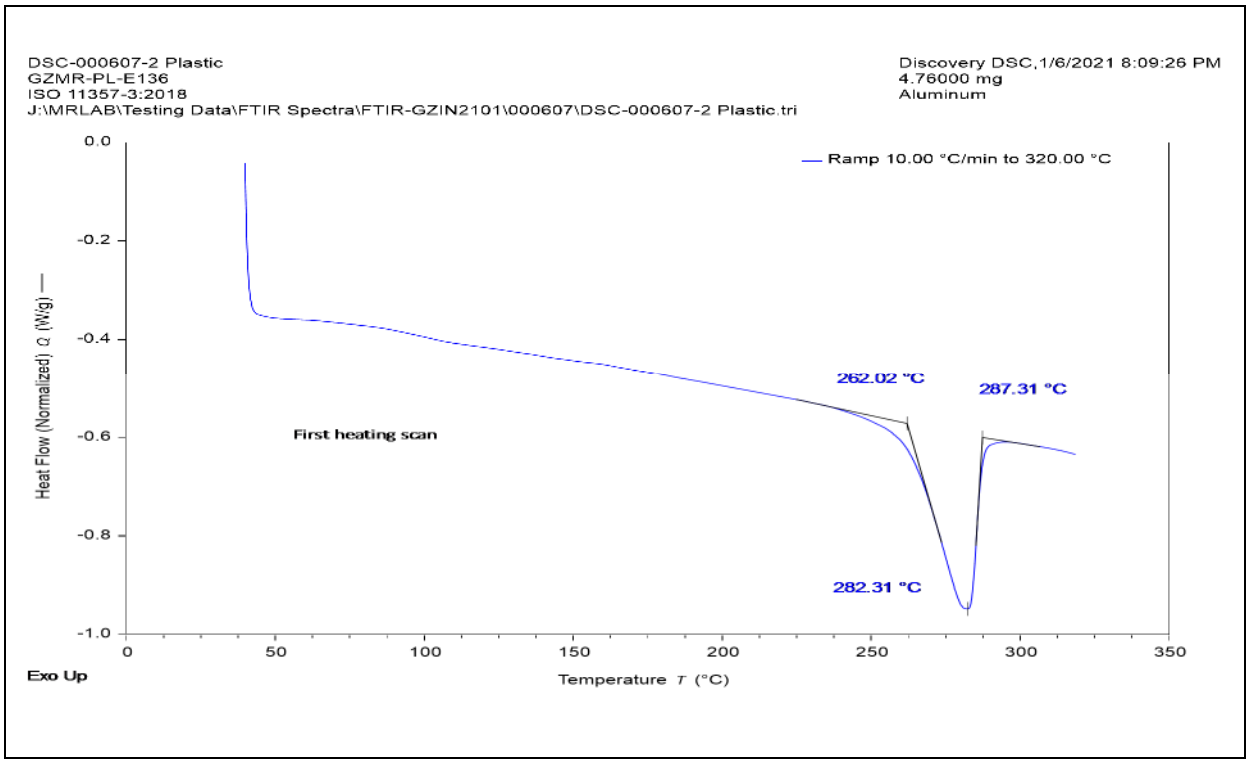
图 1



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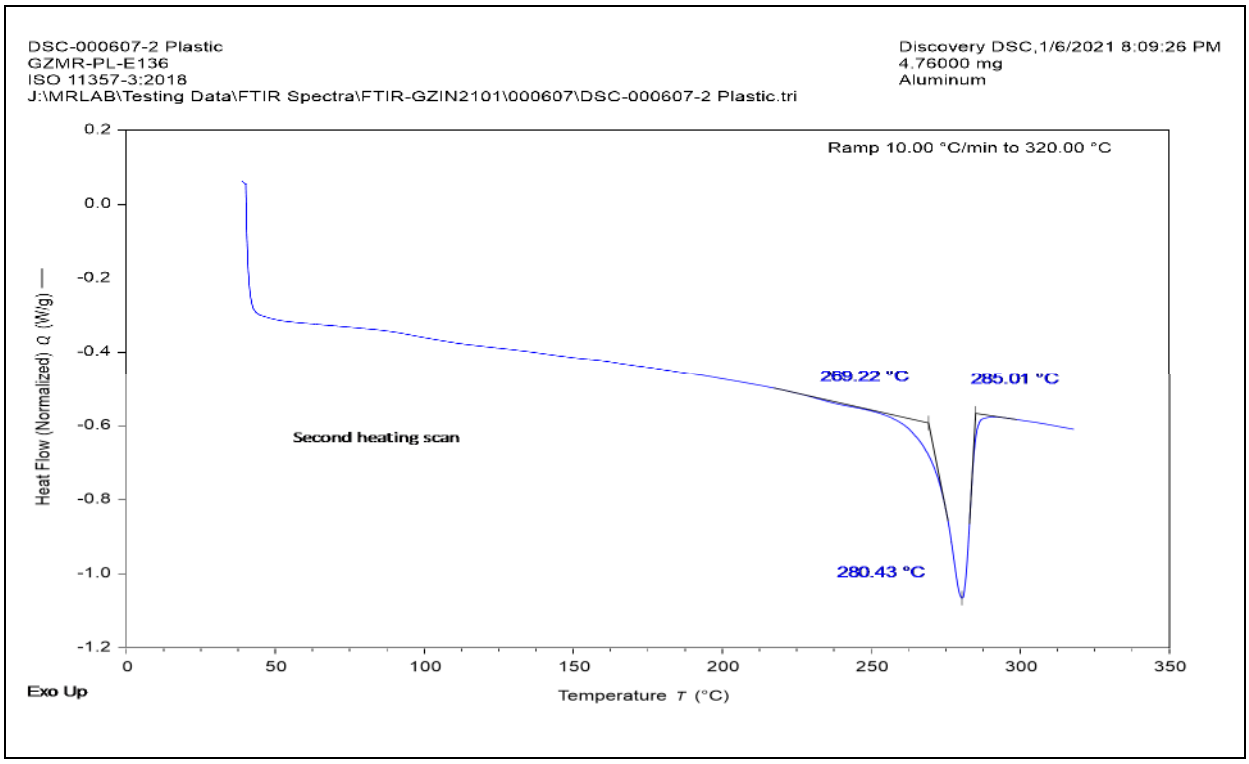
编号 ： NBIN2101000066PS_CN

日期 ： 2021-01-20

图 2



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检测报告

测试照片：

原样

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
差示扫描量热仪	Discovery DSC	GZMR-PL-E136	2020-08-11	2022-08-10

- 第 7~9 项测试由 SGS 集团内部其他实验室执行。
- 该检测报告为由 SGS 发布的原检测报告 编号：NBIN2101000066PS，日期：2021-01-20 的中文译本，本报告仅供参考。
- 在中华人民共和国境内，报告若未加盖 CMA 章，表示本检测报告仅用于客户科研、教学、内部质量控制、产品研发等目的，仅供内部参考。

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